UNIVERSITÀ di VERONA
Dipartimento di INFORMATICA

Discrete Optimization and Decision Making

Integer Linear Programming - Modeling Examples

Master's degrees in Data Science - Artificial Intelligence - Mathematics
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(Mixed) Integer Linear Programming: Modeling Examples

Selection from a set

Let us consider two continuous variables c and x. x represents the degrees of an angle $0 \leq \alpha \leq 360$. Formulate linear constraints such that α can only be a multiple of a right angle, and c takes the value of the cosine of α.

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$$\sum_{i=1}^{5} y_i = 1$$

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$$\alpha \in \{0, 90, 180, 270, 360\}$$

$$y_i \in \{0, 1\} \quad i \in \{1, 2, 3, 4, 5\}$$

$$c = 1y_1 + 0y_2 - 1y_3 + 0y_4 + 1y_5$$

A production problem 1/2

RuotaStorta S.p.A. has been involved in the production of cart wheels for years. In particular, 4 types of wheels are produced, using the only machine available. The machine, in order to produce a certain type of wheel, must be loaded with all the components first. This loading time is called *setup time* (ST). ST depends on the type of wheel that you want to produce and on which type of wheel was produced immediately before. The machine starts the day empty, and must return to this state at the end of the day. All 4 types of wheels must be produced and it is not possible to interrupt the production of one type to switch to another and then return to the production of the previous type. Formulate a **Integer Linear Programming** model which determines the **order** in which to produce the various types of wheels, **minimizing** the sum of the STs.

	Init state	Type 1	Type 2	Type 3	Type 4
Init state	-	9	-	13	-
Type 1	6	-	8	-	-
Type 2	-	-	-	-	9
Type 3	5	-	7	-	-
Type 4	-	11	-	12	-

A production problem 2/2

- $V = \{0, 1, 2, 3, 4\}$ set of wheel types (0 init state)
- A set of possible transitions (changes of production from i to j, $i, j \in V$)
- $x_{ij} \in \{0, 1\} = 1$ if type j is produced after type i, 0 otherwise
-

$$\min \sum_{(i,j) \in A} c_{ij} x_{ij} \tag{1}$$

$$\sum_{(i,j) \in \delta^-(j)} x_{ij} = 1 \qquad j \in V \tag{2}$$

$$\sum_{(i,j) \in \delta^+(i)} x_{ij} = 1 \qquad i \in V \tag{3}$$

$$u_1 = 1 \tag{4}$$

$$2 \leq u_i \leq n \qquad i \in V \setminus \{1\} \tag{5}$$

$$u_j \geq u_i + 1 - (n-1)(1 - x_{ij}) \qquad i \neq j \in V \setminus \{1\} \tag{6}$$

$$x_{ij} \geq 0 \quad \text{integer}, \quad (i, j) \in A \tag{7}$$

Organizing shifts in an hospital

At the end of the week, am hospital ward has to determine the nurses schedule for the following week. In particular, 5 nurses are part-time workers, and has to work exactly 2 days per week. To simplify schedule construction, nurses are given 3 options, and each nurse has to give each option a preference score. An option has to be chosen for each nurse, considering that each day a minimum number of nurses has to be available: 2 from monday to thursday, 1 on friday. Additionally, it is well known that nurse 1 does not want to work with nurse 4. Formulate a **Integer Linear Programming** model that maximizes the total score.

	Mon	Tue	Wed	Thu	Fri
Option 1	x			x	
Option 2			x		x
Option 3		x	x		

Table 1: Different schedule options

	Option 1	Option 2	Option 3
Nurse 1	7	5	1
Nurse 2	10	1	6
Nurse 3	6	8	4
Nurse 4	5	5	5
Nurse 5	6	2	9

Table 2: Preference scores

Notation

Nurses (i) = $1, 2, 3, 4, 5$
Options (j) = 1, 2, 3

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$$x_{ij} \in \{0, 1\} = \begin{cases} 1 & \text{if I select option j for nurse i,} \\ 0 & \text{otherwise.} \end{cases}$$

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Objective function

$$\max \quad 7x_{11} + 5x_{12} + x_{13} + 10x_{21} + x_{22} + 6x_{23} + 6x_{31} + 8x_{32} + 4x_{33} + 5x_{41} + 5x_{42}$$
$$+5x_{43} + 6x_{51} + 2x_{52} + 9x_{53}$$

One option for each nurse

$$x_{11} + x_{12} + x_{13} = 1$$
$$x_{21} + x_{22} + x_{23} = 1$$
$$x_{31} + x_{32} + x_{33} = 1$$
$$x_{41} + x_{42} + x_{43} = 1$$
$$x_{41} + x_{42} + x_{43} = 1$$
$$x_{51} + x_{52} + x_{53} = 1$$

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$$x_{51} + x_{52} + x_{53} = 1$$

At least 2 nurses every day, except for friday

$$x_{11} + x_{21} + x_{31} + x_{41} + x_{51} \geq 2$$
$$x_{13} + x_{23} + x_{33} + x_{43} + x_{53} \geq 2$$
$$x_{12} + x_{22} + x_{32} + x_{42} + x_{52} + x_{13} + x_{23} + x_{33} + x_{43} + x_{53} \geq 2$$
$$x_{12} + x_{22} + x_{32} + x_{42} + x_{52} \geq 1$$

Nurse 1 does not want to work with nurse 4

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$$x_{11} + x_{41} \leq 1$$

$$x_{13} + x_{43} \leq 1$$

$$x_{12} + x_{42} \leq 1$$

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$$x_{ij} \in \{0, 1\} \qquad \forall i \in \{1, \ldots, 5\}, j \in \{1, 2, 3\}$$

Nurse 1 does not want to work with nurse 4

$$x_{11} + x_{41} \leq 1$$

$$x_{13} + x_{43} \leq 1 \quad \text{Not needed!}$$

$$x_{12} + x_{42} \leq 1 \quad \text{Not needed!}$$

$$x_{12} + x_{42} + x_{13} + x_{43} \leq 1$$

$$x_{ij} \in \{0, 1\} \qquad \forall i \in \{1, \ldots, 5\}, j \in \{1, 2, 3\}$$

Working in a piadineria

The owner of a piadineria, needs to service the last 3 customers of the day, Patroclo, Annunziata, and Rosella. They all express their requests that they'd to be satisfied. The owner warns them that probably not all the requests will be satisfied, due to the low quantities of ingredients available. However, he ensures that that at least one ingredient will be included in each piadina. Formulate an **Integer Linear Programming model** that minimizes the number of unsatisfied requests.

Ingredient	Calories(Cal)	Category	Price(euros)	Quantity
Coppa	14	Sliced meat	2	1 piadina
Prosciutto crudo	12	Sliced meat	2.5	1 piadina
Rocket salad	5	Vegetables	1	1 piadina
Lettuce	4	Vegetables	1.5	2 piadinas
Radish	3	Vegetables	2	1 piadina
Tomatoes	3.5	Vegetables	0.5	1 piadina
Mozzarella	17	Cheese	0.5	2 piadinas
Squacquerone	25	Cheese	1.5	1 piadina

Table 3: Leftover ingredients

Customer	Cat. req.	Calories req..	Price req..
Patroclo	At least 1 sliced meat, at least 2 vegetables, at most 1 cheese	None	None
Annunziata	None	None	Max cost: 4 euros
Rosella	No cheese	At most 20 calories	None

Table 4: Requests

Notation

$C = \{P, A, R\}$ (customers)
$I = \{CO, PR, RU, L, RA, PO, M, S\}$ (ingredients)
$N_P = \{1, 2, 3\}, N_A = \{1\}, N_R = \{1, 2\}$ (requests)

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Variables

$$x_{ic} = \begin{cases} 1 & \text{if ingredient } i \text{ is used for customer } c \\ 0 & \text{otherwise.} \end{cases} \qquad \forall i \in I, c \in C$$

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$$y_{cn} = \begin{cases} 1 & \text{if request } n \text{ of customer } c \text{ is not satisfied} \\ 0 & \text{otherwise.} \end{cases} \qquad \forall c \in C, n \in N_c$$

Objective function:

$$\min \quad (y_{P1} + y_{P2} + y_{P3}) + y_{A1} + (y_{R1} + y_{R2})$$

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$$\min \quad (y_{P1} + y_{P2} + y_{P3}) + y_{A1} + (y_{R1} + y_{R2})$$

At least one ingredient for each piadina:

$$\sum_{i \in \{CO, PR, RU, L, RA, PO, M, S\}} x_{i,c} \geq 1 \qquad \forall c \in C$$

Objective function:

$$\min \quad (y_{P1} + y_{P2} + y_{P3}) + y_{A1} + (y_{R1} + y_{R2})$$

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Limit on the quantities:

$$x_{i,P} + x_{i,A} + x_{i,R} \leq 1 \qquad \forall i \in I \setminus \{L, M\}$$

$$x_{i,P} + x_{i,A} + x_{i,R} \leq 2 \qquad \forall i \in \{L, M\}$$

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$$\min \quad (y_{P1} + y_{P2} + y_{P3}) + y_{A1} + (y_{R1} + y_{R2})$$

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Patroclo's requests:

$$x_{CO,P} + x_{PR,P} \geq 1 - y_{P1}$$

$$x_{RU,P} + x_{L,P} + x_{RA,P} + x_{PO,P} \geq 2(1 - y_{P2})$$

$$x_{M,P} + x_{S,P} \leq 1 + y_{P3}$$

Annunziata's requests:

$$2x_{CO,A} + 2.5x_{PR,A} + x_{RU,A} + 1.5x_{L,A} + 2x_{RA,A} + 0.5x_{PO,A}$$

$$+0.5x_{M,A} + 1.5x_{S,A} \leq 4 + My_{A1}$$

Annunziata's requests:

$$2x_{CO,A} + 2.5x_{PR,A} + x_{RU,A} + 1.5x_{L,A} + 2x_{RA,A} + 0.5x_{PO,A}$$

$$+0.5x_{M,A} + 1.5x_{S,A} \leq 4 + My_{A1}$$

Rosella's requests:

$$x_{M,R} + x_{S,R} \leq 2y_{R1}$$

$$14x_{CO,R} + 12x_{PR,R} + 5x_{RU,R} + 4x_{L,R} + 3x_{RA,R} + 3.5x_{PO,R}$$

$$+17x_{M,R} + 25x_{S,R} \leq 20 + Ny_{R2}$$

$$x_{i,c} \in \{0,1\}, \quad \forall i \in I, c \in C$$

$$y_{c,n} \in \{0,1\}, \quad \forall c \in C, n \in N_c$$